}



Tecnológico de Monterrey

Materia:
Integración de seguridad informática en redes y sistemas de software (Gpo 601)

Actividad evaluativa:
4.2.8 Laboratorio - Configure Router-on-a-Stick Inter-VLAN Routing

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Fecha:
19 de agosto de 2026

Hard reset Switch

```
Switch#erase sta
Switch#erase startup-config
**************************************************************************************
****************************
Erasing Nvram will not clear license trust code.
**************************************************************************************
****************************
Erasing the nvram filesystem will remove all configuration files! Continue? [con
firm]
[OK]
Erase of nvram: complete
Switch#
```

```
Switch#reload

System configuration has been modified. Save? [yes/no]: no
Reload command is being issued on Active unit, this will reload the whole stack
Proceed with reload? [confirm]

*Aug 19 22:13:15.496: %SYS-5-RELOAD: Reload requested by console. Reload Reason:
 Reload Command.
Chassis 1 reloading, reason - Reload command
Aug 19 22:13:21.721: %PMAN-5-EXITACTION: C0/0: pvp: Process manager is exiting:
reload cc action requested
Aug 19 22:13:21.768: %PMAN-5-EXITACTION: F0/0: pvp: Process manager is exiting:
reload fp action requested
Aug 19 22:13:24.271: %PMAN-5-EXITACTION: R0/0: pvp: Process manager is exiting:
process exit with reload stack code



Initializing Hardware...

System Bootstrap, Version 17.9.1r [FC13], RELEASE SOFTWARE (P)
Compiled Thu 06/08/2023 20:48:41 by rel

Current ROMMON image : Primary
C9200L-24T-4G platform with 2097152 Kbytes of main memory

boot: attempting to boot from [flash:packages.conf]
boot: reading file packages.conf

##############

Both links down, not waiting for other switches
                                                Switch number is 1

              Restricted Rights Legend

Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.
```

Hard reset Router

```
Router#erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [con
firm]
[OK]
Erase of nvram: complete
Router#
*Aug 19 20:47:02.035: %SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#
Router#
```

Fecha:

```
Router#reload
Proceed with reload? [confirm]

*Aug 19 20:51:02.287: %SYS-5-RELOAD: Reload requested by console. Reload Reason: Reload Com
d.
System Bootstrap, Version 15.0(1r)M16, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2012 by cisco Systems, Inc.

Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled


Readonly ROMMON initialized
program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340


IOS Image Load Test
___________________
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x481ac1c
Self decompressing the image : ########################################################
########################################################################################
########################################################################################
########################################################################################
########################################################################################
########################################################################################
########################################################################################
######################## [OK]

Smart Init is enabled
smart init is sizing iomem
                TYPE      MEMORY_REQ
    Onboard devices &
         buffer pools      0x01E8F000
-----------------------------------------------
              TOTAL:      0x01E8F000

Rounded IOMEM up to: 32MB.
Using 6 percent iomem. [32MB/512MB]

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Rights clause at FAR sec. 52.227-19 and subparagraph
```

```
Switch>enable
Switch#confi
Switch#configure
Switch#configure t
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#vlan 10
S1(config-vlan)#name Vlan10
S1(config-vlan)#exit
S1(config)#vlan 30
S1(config-vlan)#name Vlan30
S1(config-vlan)#exit
```

```
S1#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface g1/0/6
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 10
S1(config-if)#no shut
S1(config-if)#exit
S1(config)#interface g1/0/
% Incomplete command.

S1(config)#interface g1/0/11
S1(config-if)#swi
S1(config-if)#switchport mode access
S1(config-if)#swit
S1(config-if)#switchport access vlan 30
S1(config-if)#no shut{
                     ^
% Invalid input detected at '^' marker.

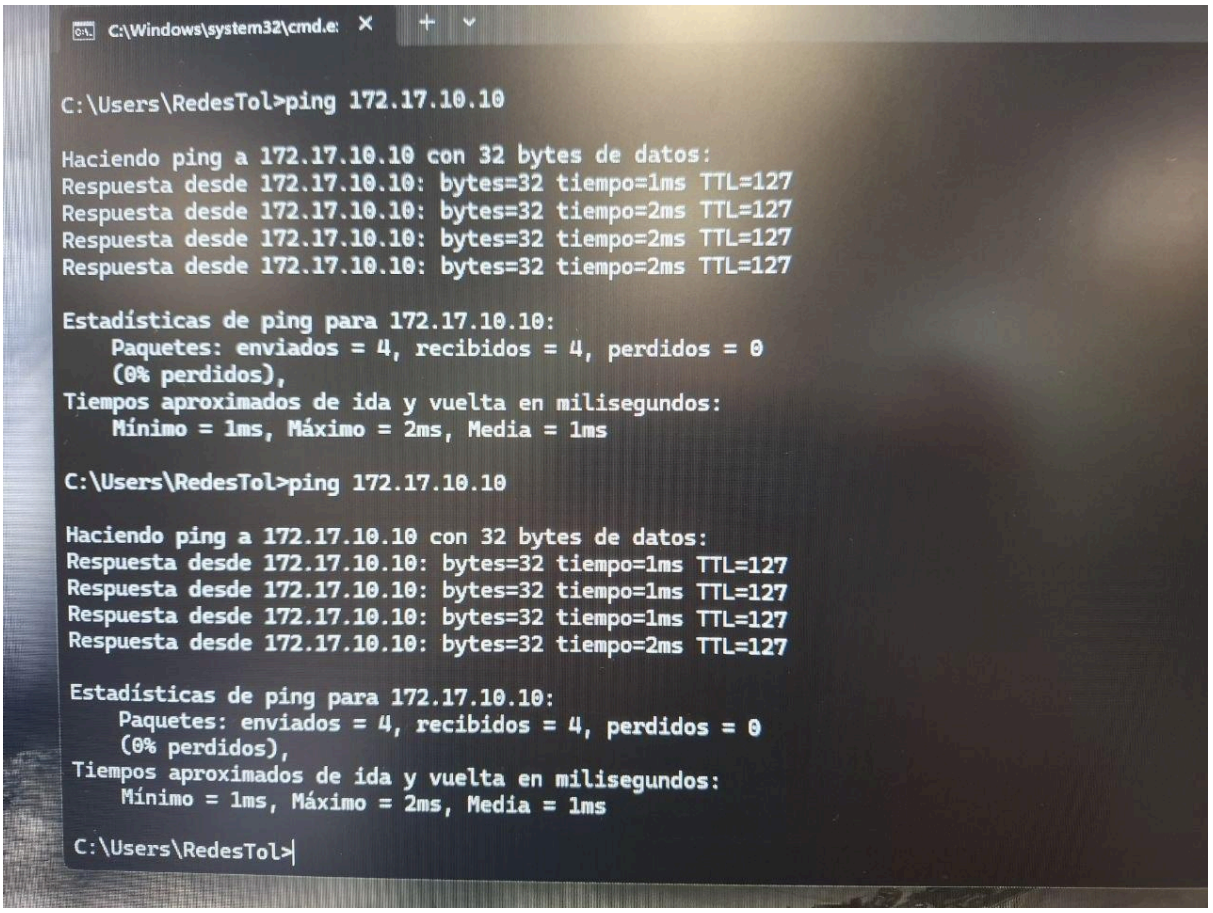
S1(config-if)#no shut
S1(config-if)#exit
S1(config)#interface g1/0/1
S1(config-if)#sw
S1(config-if)#switchport mode trunk
S1(config-if)#no shut
*Aug 19 22:57:19.388: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEth
ernet1/0/1, changed state to down
S1(config-if)#
*Aug 19 22:57:22.396: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEth
ernet1/0/1, changed state to up
S1(config-if)#
```

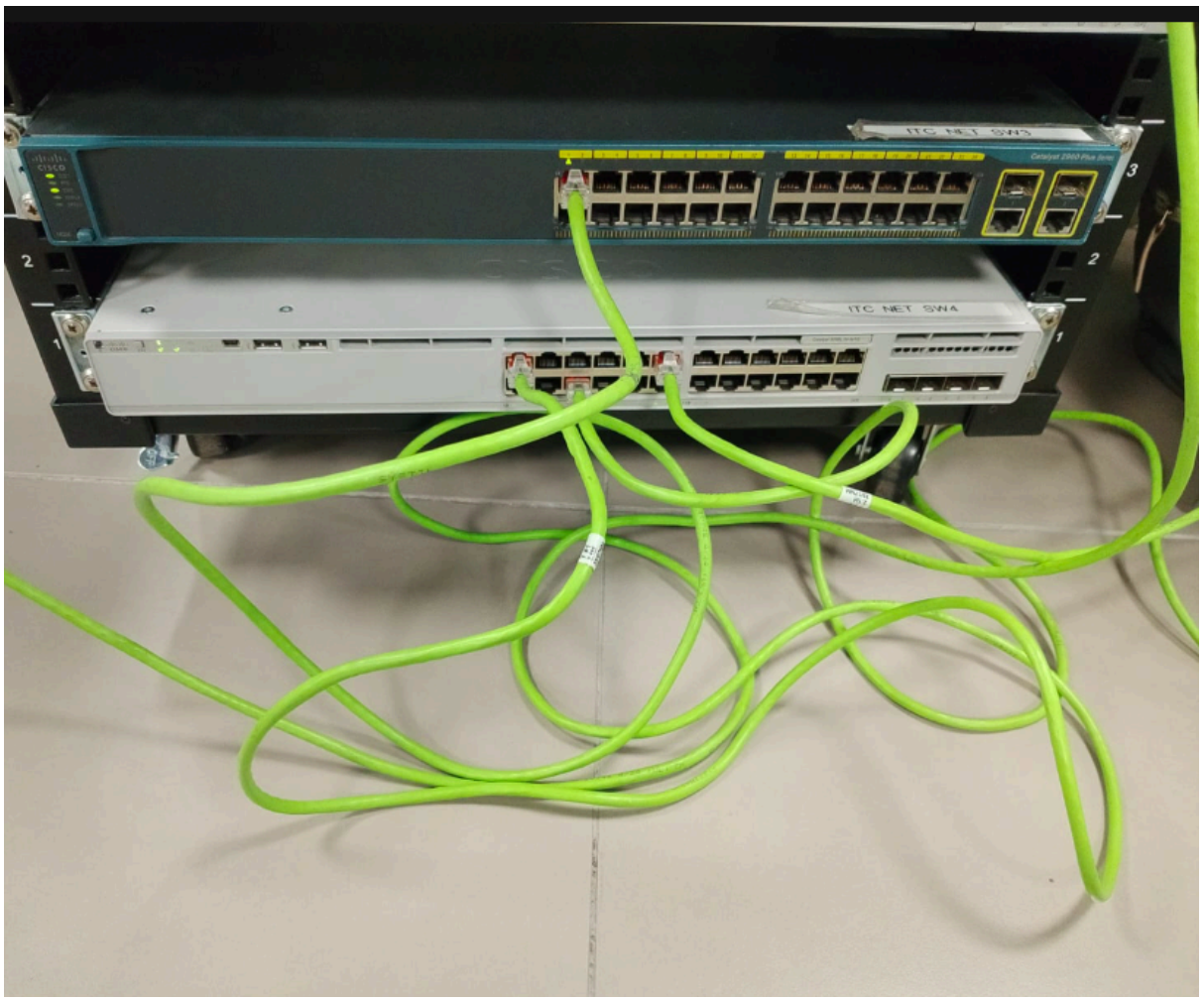
```
C:\Users\MrKan>ping 172.17.30.10

Haciendo ping a 172.17.30.10 con 32 bytes de datos:
Respuesta desde 172.17.30.10: bytes=32 tiempo=2ms TTL=127
Respuesta desde 172.17.30.10: bytes=32 tiempo=2ms TTL=127
Respuesta desde 172.17.30.10: bytes=32 tiempo=1ms TTL=127
Respuesta desde 172.17.30.10: bytes=32 tiempo=2ms TTL=127

Estadísticas de ping para 172.17.30.10:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 1ms, Máximo = 2ms, Media = 1ms

C:\Users\MrKan>
```





Paso 3: probar la conectividad entre la PC1 y la PC3. En la PC1, haga ping a la PC3.
Pregunta: ¿Fueron correctos los pings? ¿Por qué conseguiste este resultado?

Los pings no tuvieron éxito. Las PC están en diferentes redes IP y requieren un router o un switch de Capa 3 para proporcionar comunicación entre ellas.

On S1, emita el comando show vlan
¿A qué VLAN está asignado G0/1?

La vlan 1

¿Cómo puede determinar que la interfaz es un puerto de enlace troncal con el comando show vlan?

No aparece listada la interfaz

¿Qué direcciones usan PC1 y PC3 como sus direcciones de puerta de enlace predeterminadas?

172.17.10.1 y 172.17.30.1